GE POWER
PRODUCT SERVICE

TIL 2284
Publish Date: 05 MAY 2021
Compliance Category - **S**
Timing Code - **4**

TECHNICAL INFORMATION LETTER

F-CLASS BALANCE WEIGHT GROOVE ENTRY SLOT RECOMMENDATIONS

APPLICATION
6F, 7F, and 9F Heavy Duty Gas Turbines.

PURPOSE
To advise users of the recommendation to inspect compressor aft shaft (CAS) and distance piece (DP) balance weight groove entry slots corner and perform blending operations.

COMPLIANCE CATEGORY

M - Maintenance	Identifies maintenance guidelines or best practices for reliable equipment operation.
C - Compliance Required	Identifies the need for action to correct a condition that, if left uncorrected, may result in reduced equipment reliability or efficiency. Compliance may be required within a specific operating time.
A - Alert	Failure to comply with the TIL could result in equipment damage or facility damage. Compliance is mandated within a specific operating time.
S - Safety	Failure to comply with this TIL could result in personal injury. Compliance is mandated within a specific operating time.

TIMING CODE

1	Prior to Unit Startup / Prior to Continued Operation (forced outage condition)
2	At First Opportunity (next shutdown)
3	Prior to Operation of Affected System
4	At First Exposure of Component
5	At Scheduled Component Part Repair or Replacement.
6	Next Scheduled Outage

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BACKGROUND DISCUSSION

A gas turbine rotor is an assembly of shafts, wheels, disks, and spacers. A means of balance is incorporated into each of these rotor components and the various assemblies. A balance weight groove (BWG) is one example of a feature that is used to provide a means of balancing. The BWG is an annular channel that is machined into a rotor component that provides a location for weights to be installed for balance purposes on either the piece part or as a rotor assembly. This annular channel may contain two BWG entry slots, as shown in **Figure 1**, located approximately 180 degrees apart that allow for the balance weights to be easily installed and located in the appropriate locations on the wheel or shaft.



Figure 1: BWG Entry Slots - Correct Machining

During non-destructive evaluation (NDE) of a distance piece (DP) as part of a rotor life extension (RLE), a crack-like indication was observed in the corner of a balance weight groove entry slot. Both grooves on the specific part showed similar indications as depicted in **Figure 2**. GE has inspected additional parts with similar features and observed similar indication findings.

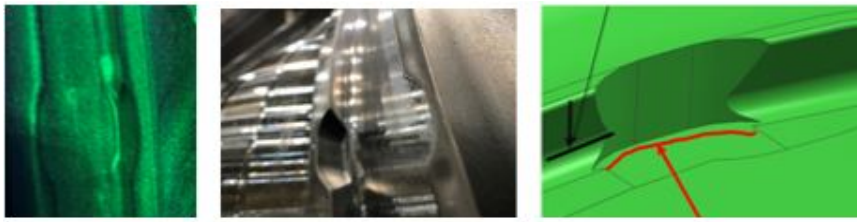


Figure 2: Indications Found on DP During RLE - Observed Machining Issue

Upon completion of a root cause investigation, GE determined the cause is associated with the variation in the corner condition of the balance weight groove entry slots and interactions with the various stresses during gas turbine start-up, shutdown, and loaded operation (gravity sag, etc). Depending upon the specific nature of a units operation and the specific condition of the part slot corner, crack-like indications may form over the lifetime of the component. If these crack-like indications propagate to a large enough size, it may be possible for continued rapid crack growth to occur associated with gravity sag related bending of the rotor as it rotates (high cycle fatigue related mechanism). If not mitigated, the fracture propagation may lead the CAS and/or DP to separate into two separate pieces. Separation of either shaft into two pieces may create the possibility for components to exit the gas turbine leading to the possibility for INJURY TO PERSONNEL/INDIVIDUALS and substantial damage to adjacent equipment.

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RECOMMENDATIONS

It is recommended to perform fluorescent penetrant inspection (FPI) for all BWG entry slots on the CAS and DP during the next planned outage for the unit Hot Gas Path (HGP) or Major Inspection (MI) at or before the rotor reaches a total of 378,000 equivalent cold rotor turning time.

For case where FPI is clear of any indications – evaluate all the balance entry slot corner conditions with replication process.

- If any balance weight entry slot corner condition presents with a <0.060” fillet size - perform entry slot blending, FPI, and peening operations to mitigate the risk of crack initiation and propagation.
    - Corner specific item - if any portion of a specific balance weight groove corner meets noted criteria, then perform modification machining on the entire balance weight groove entry slot corner.
- Fillet > 0.060" and uniformly this size over full length of the BWG entry slot corner – leave in as-found condition

For case with observable FPI indication/indications - contact GE engineering through an ER case for technical disposition.

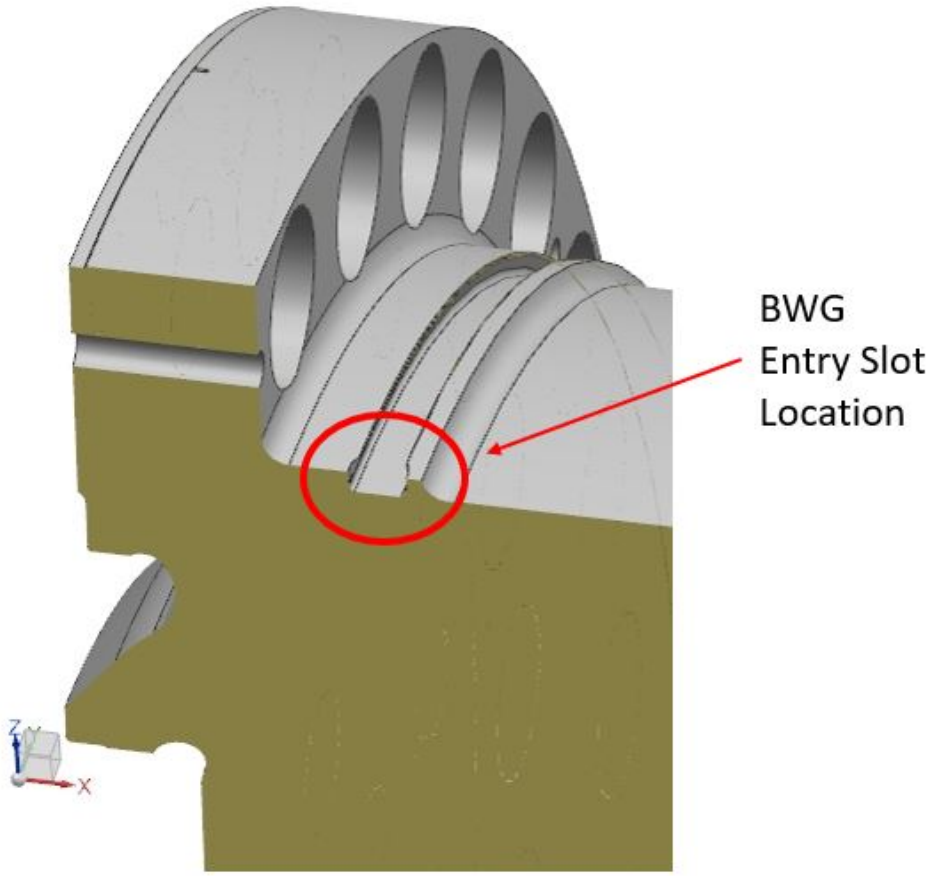


Figure 3: Relative Location of BWG Entry Slots

Equivalent cold rotor turning time is defined as a spinning rotor in combination with cool metal temperatures (generally speaking conditions with all wheelspace temperatures measuring less than 150 degrees F). Turning gear (TG) operation, forced cooling operation, offline water wash operation, and unit startups are the periods where cool rotor conditions exist and contribute to the overall equivalent cold rotor turning time. Equivalent refers to a standardization of the # of high cycle fatigue cycles for a given hour and therefore has a component associated with the accumulation of cycles relative to the rotor speed. For example, at purge speed (~20% speed) versus turning gear speed (~6 rpm) a higher # of stress cycles will occur per minute of operation due solely to the increased rotational speed of the rotor for a given period.

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Therefore, the equivalent cold rotor turning time impact is based around taking all cold rotational hours and converting it into an equivalent number of turning hours at typical turning gear speed. In order to calculate equivalent cold rotor turning time, GE recommends the following methods:

Equivalent Cold Rotor Turning = ECRT (hours)

$$ECRT = T_{TG}-((S_{FS}-S_{FC})*X) + T_{FC}*0.25*\frac{20\%\ Full\ Speed}{6} + (S_{WC}*Y) + (S_{WW}*3*\frac{20\%\ Full\ Speed}{6})$$

Path #1 (Unit historical recorded data through GE OSM connection or customer records)

- Record the # of turning gear hours (some controllers display this value) - T_{TG}
- Record the total # of fired starts – S_{FS}
    - For 7FAs – X = 24
    - For 9FAs – X = 24
    - For 6FAs – X = 18
- Record # of force cool downs – S_{FC}
- Record # of force cool down hours – T_{FC}
- Unit Full Speed
    - For 7FAs – 3600 rpm
    - For 9FAs – 3000 rpm
    - For 6FAs – 5250 rpm
- Record the # of cooler starts unit has completed (defined as the # of warm 2 and cold starts per the definitions in GER-3620 (latest revision – definition is provided in figure 22). - S_{WC}
    - Start Up Factor
        - Simple Cycle Unit (SC) Y = 72
        - Combined Cycle Unit (CC) Y = 86
- Record # of offline water wash operations – S_{WW}

Note: If unit has greater than 50% of their operation covered by On-Site Monitoring (OSM) connection recorded data, then ratio data up to cover 100% of operation with this data, if less than 50% utilize customer records or memory to determine if 50% covered by OSM connected data is appropriate to cover overall unit operation.

Path #2 (if do not have data available for path #1):

- Record the # of turning gear hours (some controllers display this value) - T_{TG}
- Record the total # of fired starts – S_{FS}
    - For 7FAs – X = 24
    - For 9FAs – X = 24
    - For 6FAs – X = 18
- Utilize average % of cooler starts value assumption - $S_{WC}= S_{FS}*0.72$
    - Start Up Factor
        - Simple Cycle Unit Y = 72
        - Combined Cycle Unit Y = 86

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- Utilize average % of force cool downs value assumption – T_{FC}= S_{FS}*0.05*Z
  - For 7FAs – Z = 8
  - For 9FAs – Z = 12
  - For 6FAs – Z = 24

Example ECRT Calculation:

Combined Cycle 7FA.03

T_{TG}	S_{FS}	S_{FC}	X	T_{FC}	S_{Wc}	Y	S_{WW}
Turning Gear Time (hours)	Number of Fired Starts	Number of Forced Cool Downs	Frame Size Factor	Forced Cool Time (hours)	Number of Cool Starts	SC or CC Factor	Number of Offline Water Washes
40,000	1,900	240	24	4800	1300	86	40

ECRT = 40,000 – (1,900-240) *24 + 4800*0.25* $\frac{20\%\ of\ 3600}{6}$ + (1300*86) +(40*3* $\frac{20\%\ of\ 3600}{6}$)

ECRT = 270,360 hours

This TIL may be considered complete after completion of the FPI/MPI inspection and the CAS and DP geometry is modified. Please contact your GE Service Representative for any questions associated with the implementation of this TIL.

PLANNING INFORMATION

Compliance

• Compliance Category: **S**

• Timing Code: **4**

Manpower Skills

- GE qualified specialist to perform FPI (in service center can substitute MPI inspection for FPI inspection based on service center capability)
- GE qualified machinist with appropriate tooling to perform blending and peening operations

Special Tooling

- BWG Entry Slot Portable Machining Tool
- FPI inspection capability
- Flapper peening kit

Scope of Work

- FPI to inspect BWG entry slot pre-blending condition
- Set up BWG Entry Slot tool and perform blending operation
- FPI to inspect BWG entry slot post-blending condition
- Flapper peen the modified geometry post machining and post machining successful FPI

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TIL DISPOSITION

Disposition of TILs should be entered in local records and also in GE Power ServiceNow. Follow the below instructions for entering the disposition record;

- Log into the Power ServiceNow at https://gepowerpac.service-now.com/til_new/ using your GE SSO number and password.
- Select "TIL Disposition".
- Click on the TIL for the serial number you want to update.
- Choose the most appropriate "Disposition Status" and enter "Disposition Notes".
- Click "Save".

Contact your local GE Services representative for assistance or for additional information

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